GGG-B-211b
April 5, 1967
SUPERSEDING
Fed. Spec. GGG-B-211a
December 4, 1953

FEDERAL SPECIFICATION

BEVELS: SLIDING T; UNIVERSAL: AND SHIP CARPENTERS'

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bevels used by machinists and carpenters for measuring and transferring angles in layout work.

1.1.1 *Federal specification coverage.* This specification does not cover all types and classes of bevels commercially available, but only those generally used by the Federal Government.

1.2 Classification. Bevels shall be of the following types and classes, as specified (see 6.2):

Type I—Sliding T.
Class 1—Wood handle.
Class 2—Metal handle.
Type II—Universal.
Type III—Ships carpenters'.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standards:

MIL-STD-105—Sampling procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of bevels which are otherwise in accordance with the requirements of this specification.

3.2 Type I, sliding T. Type I bevels shall be of two classes and each shall consist of a

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TABLE I. *Type I, class 1, wood handle*

Size	Blade			Handle		
	Length ± 1/4	Width ± 1/16	Thickness ± 0.005	Length ± 1/8	Width ± 1/16	Thickness ± 1/16
	Inches	*Inches*	*Inch*	*Inches*	*Inch*	*Inch*
6	6	7/8	0.050	4-1/2	7/8	23/32
8	8	1	.050	5-9/16	1	23/32
10	10	1-1/16	.050	6-9/16	1-1/16	23/32
12	12	1-1/16	.050	7-9/16	1-1/16	23/32

blade and handle with a locking device to hold the blade in any position relative to the handle. Class 1 shall have a wood handle and class 2 shall be provided with a metal handle.

3.2.1 *Type I, class* 1. Type I, class 1 bevels shall be similar to figure 1, comply with the requirements of table I, and shall have a handle of straight grained hardwood, bound at each end with metallic tips securely fastened to the wood handle. The tips shall be made of a corrosion-resisting metal or steel treated to resist corrosion. The wooden handle shall be smoothly finished and treated to resist absorption of moisture. The blade shall be furnished in lengths of 6-, 8-, 10-, and 12-inches, as specified (see 6.2). The blade shall be blued, nickel plated, or stainless steel. Blades shall have a minimum hardness of 90 on the Rockwell "B" scale.

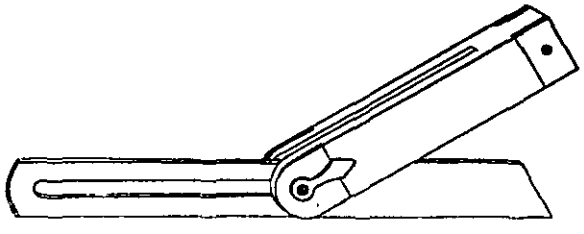


FIGURE 1. Type I, class 1, sliding T, wood handle.

3.2.2 *Type I, class* 2. Class 2 bevels shall be similar to figure 2. The blade and handle shall be of ground steel with a hardness of not less than 90 on the Rockwell "B" scale. The blade thickness shall be 0.050 ± 0.005 inch and its length shall be approximately 6 ± 1/4 inch. The blade shall have two slots, one approximately 4 inches long and the other 1-1/2 inches long. The handle length shall be 3-1/2 ± 1/4 inch, its width shall be 3/4 ± 1/8 inch, and its thickness shall be 13/64 ± 1/32 inch. The surface roughness of the utile surfaces shall not exceed 32 microinches roughness height rating.

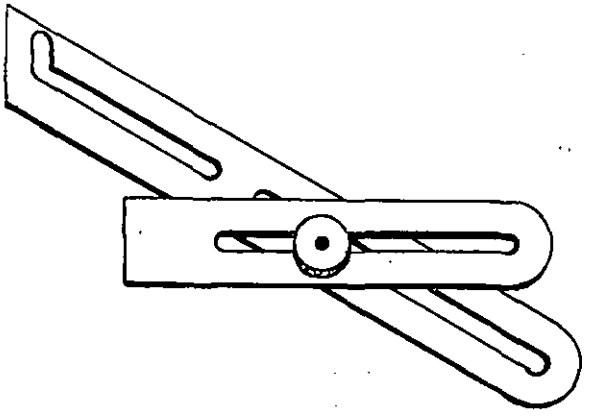


FIGURE 2. Type I, class 2, sliding T bevel, metal handle.

3.3 **Type II.** Type II bevels shall be similar to figure 3 and shall meet all requirements for type I, class 2, except for size and shape of blade. The blade shall be 0.040 ± 0.005 inch thick, and shall form a right angle with a center longitudinal slot extending into both legs. One leg shall have a beveled end shall be 3 ± 1/4 inch in length, and the other leg shall be 1-5/8 ± 1/4 inch in length. The surface finish of the bevel shall not exceed 32 microinches roughness height rating. The handle length shall be 3 ± 1/4 inch, its width shall be 11/16 ± 1/8 inch, and its thickness shall be 5/32 ± 1/32 inch. The blade and handle shall have a minimum hardness of 90 on the Rockwell "B" scale.

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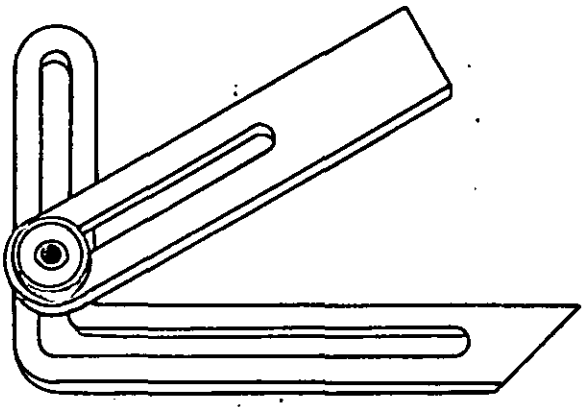


FIGURE 3. Type II, universal bevel.

3.4 Type III, ships carpenters'. Type III bevels shall consist of a body and two blades. The blades shall be hinged one to each end of the body and be similar to figure 4.

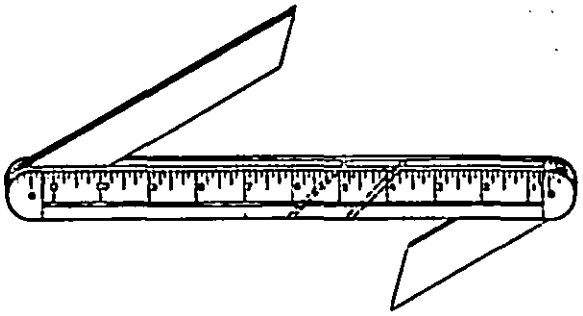


FIGURE 4. Type III, ships carpenters' bevel.

3.4.1 *Body.* The body shall be of birchwood, boxwood or maple wood, and shall be 12 ± 1/4 inch in length, designed to encase both blades when folded. The body faces shall be graduated in inches and eights of an inch and the ends shall be bound with metallic tips. The wood shall be smoothly finished and treated to resist the absorption of moisture. The tips shall be either corrosion-resisting metal or steel treated to resist corrosion.

3.4.2 *Blades.* The blades shall be of brass or steel treated to resist corrosion and shall pivot at each end of the body.

3.5 Surface finish. Burrs shall be removed from all metal edges and sharp fins shall be removed from all wood edges. All surfaces used for layout work shall be straight and smooth. Metallic tips shall be fitted flush with the wood surfaces upon which they are fastened.

3.6 Marking.

3.7 Workmanship. Workmanship of the bevels shall be of the quality prevailing among manufacturers normally producing bevels of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling Procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table II.

4.3 Examination.

4.3.1 *Visual examination.* Each sample unit shall be examined for any nonconformance in design, material, finish, coating, workmanship, and marking. Defects are listed in table III.

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TABLE II. *Sampling data*

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1 ea.	II	Major 2.5 minor 4.0	Percent defective	4.3.1
Dimensional examination	1 ea.	II	2.5	Percent defective	4.3.2
Testing	1 ea.	II	2.5	Percent defective	4.4
Preparation for delivery	One container	S-2	4.0	Defects per hundred units	4.5

TABLE III. *Classification of defects*

Categories	Defects
Critical	None defined.
Major:	
101	Body or handle not free from chips, burrs, or sharp edges or splinters (as applicable).
102	Type and class not as specified.
103	Material not of the type specified.
104	Incomplete; parts missing.
105	Blade or blades damaged; not the correct shape.
106	Wooden handle not free from knots, splits, or cracks; not smoothly finished (as applicable).
107	Graduations not in inches and eighths of an inch; missing or illegible (type III).
108	Body fails to enclose blades when folded (type III).
109	Body end of wooden handle not bound with metallic tips (as applicable).
110	Blades fail to slide or pivot freely or excessive tightness.
111	Wooden parts not treated to resist moisture absorption.
112	Locking device not provided or fails to properly lock in position (types I and II).
Minor:	
201	Marking not permanent, illegible, or incorrect.
202	Packaging and packing not as specified.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 **Testing.** Each sample unit shall be tested in accordance with 4.4.1 and 4.4.2.

4.4.1 *Surface finish test. The metal blades* and bodies of type I, class 1, and type II bevels shall be tested to determine compliance with the surface finish requirements specified herein.

4.4.2 *Hardness test.* The blades of types I and II bevels shall be tested to verify compliance with the hardness requirements specified herein.

4.5 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table IV. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE IV. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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TABLE V. *Preservation and quantity per unit and intermediate package and shipping container*

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Bevel:						
Sliding T						
Wood handle	P-8	IC-1	6	—	8	48
Metal handle	P-8	IC-1	6	—	8	48
Universal	P-8	IC-1	6	—	8	48
Ships carpenters'	P-8	IC-1	6	—	6	36

5. PREPARATION FOR DELIVERY

5.1 **Preservation, packaging, and packing.** Unless otherwise specified (see 6.2), the bevels shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation and packaging shall be A or C and the level of packing A, B, or C, as specified (see 6.2).

5.1.1 As part of the requirements of section 5 of this specification, table V is considered as part of and used in lieu of that portion of table I of MIL-H-15424 covering bevel, carpenter, sliding universal.

5.2 **Marking.**

5.2.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Bevels covered by this specification are generally used for layout work or transferring angular layouts from one work, location or workpiece to another.

6.2 **Ordering data.** Purchasers should select the preferred options offered herein and include the following data in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of bevel required (see 1.2).
(*c*) Blade length for type I, class 1 bevels (see 3.2.1).
(*d*) Levels of preservation, packaging, and packing required (see 5.1).

MILITARY INTEREST:
CUSTODIANS:
Army—MO
Navy—SH
Air Force—67

Review activities:
Army—GL, MO
Navy—SH, YD
Air Force—67

User activities:
Army—MO
Navy—MC—SH
Air Force—67
GSA—FSS

Preparing activity:
Navy—SH

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